Tugas 8

1) a) metode trapesium $a=0$ $b=6$ $n=6$ $h=\frac{b-a}{n}=\frac{6-0}{6}=1$

$JT=\frac{h}{2}[f_0+2f_1+2f_2+2f_3+2f_4+2f_5+f_6]$

$JT=\frac{1}{2}[0+2(3,2)+2(5,1)+2(6)+2(5,4)+2(3,1)+0,9]$

$JT=\frac{1}{2}[0+6,4+10,2+12+10,8+6,2+0,9]$

$JT=\frac{1}{2}(46)$

$JT=23$ kN·s

b) metode simpson 1/3

$JS=\frac{h}{3}[f_0+4f_1+2f_2+4f_3+2f_4+4f_5+f_6]$

$JS=\frac{1}{3}[0+4(3,2)+2(5,1)+4(6)+2(5,4)+4(3,1)+0,9]$

$JS=\frac{1}{3}[0+12,8+10,2+24+10,8+12,4+0,9]$

$JS=\frac{1}{3}[70,6]$

$JS=23,5333$ kN·s

c) ubah implus dari kN·s menjadi N·s

1kN·s = 1000 N·s

$JT=23$ kN·s $\quad JS=23,5333$ kN·s

$=23000$ N·s $\quad =23533,3$ N·s

d) menghitung perubahan kecepatan ($\Delta v$)

$m=50$ kg Trapesium Simpson 1/3

$\Delta v=\frac{J}{m}$ $\quad \Delta v_T=\frac{23000}{50}$ $\quad \Delta v_S=\frac{23533,3}{50}$

$=460$ m/s $\quad =470,666$ m/s

e) menghitung presentase selisih relatif

Rumus selisih relatif $=\frac{|JS-JT|}{JS}\times 100\%$

$=\frac{|23,5333-23|}{23,5333}\times 100\%$

$=\frac{0,5333}{23,5333}\times 100\%$

$=0,02266\times 100\%$

$=2,266\%$

2) a) menghitung $\frac{du}{dx}$ di $x = 0{,}0$ mm dgn selisih pusat

Rhumus selisih pusat  $h = 0{,}1$

$$\left(\frac{du}{dx}\right)_i = \frac{U_{i+1} - U_{i-1}}{2h}$$

$x=0$  $U(0{,}1) = 0{,}060$

$U(-0{,}1) = 0{,}055$

Subt: $$\frac{du}{dx} = \frac{0{,}060 - 0{,}055}{2(0{,}1)}$$

$$= \frac{0{,}005}{0{,}2}$$

$$= 0{,}025 \ \mu J/mm$$

b) menentukan gaya $F(0)$

hub gaya dan energi potensial

$$F(x) = -\frac{du}{dx}$$

$$F(0) = -(0{,}025)$$

$$= -0{,}025 \ mN.$$

c) menghitung $\frac{d^2u}{dx^2}$ di $x = 0$ mm.

rumus selisih pusat orde 2

$$\left(\frac{d^2u}{dx^2}\right)_i = \frac{U_{i+1} - 2U_i + U_{i-1}}{h^2}$$

$U(0{,}1) = 0{,}060$

$U(0{,}0) = 0{,}00$

$U(-0{,}1) = 0{,}055$

$$\frac{d^2u}{dx^2} = \frac{0{,}060 \pm 2(0) + 0{,}055}{(0{,}1)^2}$$

$$= 11{,}5 \ \mu J/mm^2$$

d) menghitung $\frac{du}{dx}$ dan $\frac{d^2u}{dx^2}$ di $x = -0{,}3$

selisih maju

Turunan pertama:

rumus selisih maju: $\frac{du}{dx} = \frac{U_{i+1} - U_i}{h}$

$U(-0{,}3) = 0{,}500$

$U(-0{,}2) = 0{,}220$

$$= \frac{0{,}1220 - 0{,}500}{0{,}1}$$

$$= \frac{-0{,}1280}{0{,}1}$$

$$= -2{,}8 \ \mu J/mm$$

Turunan kedua

$$\frac{d^2u}{dx^2} = \frac{U_{i+2} - 2U_{i+1} + U_i}{h^2}$$

$U(-0{,}1) = 0{,}055$

$U(-0{,}2) = 0{,}220$

$U(-0{,}3) = 0{,}500$

$$\frac{d^2u}{dx^2} = \frac{0{,}055 - 2(0{,}220) + 0{,}500}{(0{,}1)^2}$$

$$= \frac{0{,}115}{0{,}01} = 11{,}5 \ \mu J/mm^2$$

e) menghitung $\frac{du}{dx}$ dan $\frac{d^2u}{dx^2}$ di $x = 0{,}3$ mm

selisih mundur

Turunan pertama

$U(0{,}3) = 0{,}590$

$U(0{,}2) = 0{,}250$

$U(0{,}1) = 0{,}060$

$$\frac{du}{dx} = \frac{U_i - U_{i-1}}{h}$$

$$= \frac{0{,}590 - 0{,}250}{0{,}1}$$

$$= \frac{0{,}340}{0{,}1} = 3{,}4 \ \mu J/mm$$

Turunan kedua

$$\frac{d^2v}{dx^2} = \frac{v_i - 2v_i - + v_{i-2}}{h^2}$$

$$= \frac{0{,}590 - 2(0{,}250) + 0{,}060}{(0{,}1)^2}$$

$$= \frac{0{,}150}{0{,}01} = 15 \text{ NJ/mm}^2$$

f) komentar stabilitas titik $x=0$

$$\frac{d^2v}{dx^2} = 11{,}5 \text{ NJ/mm}^2 > 0$$

kurva cekung ke atas

(minimum lokal)

$\frac{dv}{dx} = 0{,}025$ NJ/mm nilai dekat dengan nol sehingga titik $x=0$ dapat dianggap berada di sekitar titik minimum energi potensial

3) $\frac{dv}{dt} = 9{,}81 - 0{,}08v^2$

$g = 9{,}81 \text{ m/s}^2$ $t_0 = 0$ $t_3 = 1{,}5$

$\beta = 0{,}08 \text{ m}^{-1}$ $t_1 = 0{,}5$ $t_4 = 2$

$h = 0{,}5$ s $t_2 = 1$

fungsi $f(t,v) = 9{,}81 - 0{,}08v^2$

a) metode euler

$$v_{n+1} = v_n + hf(t_n, v_n)$$

$$= v_n + 0{,}5(9{,}81 - 0{,}08v_n^2)$$

iterasi 1 $t_0 = 0$ $v_0 = 0$

$F(t_0, v_0) = 9{,}81 - 0{,}08(0)^2$

$= 9{,}81$ m

$v_1 = 4{,}905$ m/s

iterasi 2 $t_1 = 0{,}5$ $v_1 = 4{,}905$

$F(t_1, v_1) = 9{,}81 - 0{,}08(4{,}905)^2$

$= 9{,}81 - 1{,}924722$

$= 7{,}885278$

$v_2 = 4{,}905 + 0{,}5(7{,}885278)$

$= 4{,}905 + 3{,}942639$

$= 8{,}847639$ m/s

iterasi 3

$F(t_2, v_2) = 9{,}81 - 0{,}08(8{,}847639)^2$

$= 9{,}81 - 6{,}2624457$

$= 3{,}547543$

$v_3 = 8{,}847639 + 0{,}5(3{,}547543)$

$= 10{,}621411$ m/s

iterasi 4

$v_3 = 10{,}621411$

$F(t_3, v_3) = 9{,}81 - 0{,}08(10{,}621411)^2$

$= 9{,}81 - 9{,}025230$

$= 0{,}784770$

$v_4 = 10{,}621411 + 0{,}5(0{,}784770)$

$= 11{,}013769$

$v(2{,}0) = 11{,}013769$ m/s

b) metode Runge kutta orde 4

rumus $k_1 = f(t_n, v_n)$

$k_2 = f(t_n + \frac{h}{2}, v_n + \frac{h}{2} k_1)$

$k_3 = f(t_n + \frac{h}{2}, v_n + \frac{h}{2} k_2)$

$k_4 = f(t_n + h, v_n + h k_3)$

$v_{n+1} = v_n \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$\frac{h}{2} = 0{,}25$

$\frac{h}{6} = 0{,}083333$

langkah 1 (0-0,5)

$v_0 = 0$

$k_1 = 9{,}81 - 0{,}08(0)^2$

$= 9{,}81$

$k_2 = 9{,}81 - 0{,}08(0 + 0{,}25 \cdot 9{,}81)^2$

$= 9{,}81 - 0{,}481180$

$= 9{,}328820$

$k_3 = 9{,}81 - 0{,}08(0 + 0{,}25 \cdot 9{,}32882)^2$

$= 9{,}81 - 0{,}08(5{,}439100)$

$= 9{,}374866$

$k_4 = 9{,}81 - 0{,}08(0 + 0{,}5 \cdot 9{,}374860)^2$

$= 9{,}81 - 0{,}08(21{,}972028)$

$= 8{,}052238$

$v_1 = 0 + \frac{0{,}5}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$= 0{,}083337\ (9{,}81 + 18{,}65769 + 18{,}74973 + 8{,}05224)$

$= 0{,}083331\ (55{,}26961)$

$= 4{,}160580$ m/s

Perbandingan hasil

| | |
|---|---|
| euler | 11,0138 |
| RK4 | 10,4425 |

selisih 11,01738 - 10,4425

= 0,5713 m/s